

Part II.)

Background

Principle Component Analysis (PCA) detects the main patterns of can be used to detect facial features from a data set of random faces. Of the dataset, each face image was reshaped from a matrix into a long column vector, which were then placed into a large matrix A , in which each column represented features of one face.

The average of each column of A was then computed, giving the “mean” face, which was then subtracted from each image, producing a centered matrix (mean centreing). Mean centreing was important since PCA only investigates how each image differs from the average, while not necessarily considering the brightness or other common facial features.

The singular value decomposition (SVD) was then applied to the centred matrix A , given by:

$$A_c = U\Sigma V^T$$

Each parts of the matrix of the SVD form represented different things:

- Columns of U represented the eigenfaces, which are main patterns of variations in the image set.
- The largest singular values correspond to the most important eigenfaces, capturing the strongest patterns PCA does not capture, such as: brightness, shadows and most importantly: facial structures, which is used for the detection of moustaches as required specifically for the task.

The moustache detector involved projecting each centred image onto a chosen eigen face, producing a score measuring how strongly the image matches the moustache pattern. A threshold was then calculated from labelled moustache to non-moustache examples. Image are then classified depending on which side of the threshold their score lies.

Results

Through the development of a model on MATLAB, a moustache detector with 64.1% accuracy. When analysing the 30 random faces MATLAB used for moustache detection. The results appeared to be reasonable. With images that had a highly visible moustache, to the human eye, MATLAB seemed to have detected a moustache too. Indicating, as long as a moustache was visible, MATLAB would be able to detect it too. However, the detector was not completely reliable, some images without moustaches were classified as having moustaches; likely due to strong shadows around the mouth or nose, indicating imperfection, suggesting detector was not detecting image patterns similar to moustache

eigenfaces. These patterns included dark regions near the upper lip, shadows cause by lighting (Image 140, Figure 1)

Figure 1. detection random 30 faces.

